RESEARCH ARTICLE AND REPRODUCIBILITY ARCHIVE

Smaller explicit Diophantine equations for ZFC consistency

Encoding only the monomials that occur, and recovering a deleted Pell quotient

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Abstract

We construct explicit integer-coefficient polynomials whose absence of zeros, on precisely specified domains, is equivalent to syntactic consistency of ZFC through the repository's calibrated blank-K215 compiler. The comparison includes the eight wave-6 reports at the pinned revision, not just the older collection summary. A support-dependent quadratic encoding lowers the nine-positive-variable degree from 129 875 980 to 30 947 584, with 651 binary arithmetic operations instead of 788. Its ten-unrestricted-integer companion has degree 56 593 864 and 653 operations, versus 237 520 456 and 782. The general ten-integer reduction is Sun's; the contribution is the smaller explicit source encoding and its specialization.

A separate construction uses eleven positive variables, degree 954, and 115 operations. It combines a refined auxiliary-root estimate with a midpoint carrier and the modulus $4q^4$, which recovers the deleted index quotient without a translation of the scale variable. A lower-degree fourteen-positive-variable alternative retains degree 56 with 79 operations, removing one variable from the historical degree-56 construction. All constants are fixed by finite recipes and binary payloads; no consistency-dependent choice or unspecified enumeration index occurs. The archive contains the literal circuits, complete new reduction proofs, exact finite certificates, and preserved source notices. The inherited compiler calibration and number-theoretic interfaces are identified explicitly. This is an ordinary mathematical construction, not a new end-to-end Lean certificate or a proof of global minimality.

Reviewed repository revision: `124c07c0b77a9e20e84676514d8b1ff78536ea3f`

https://github.com/VladimirReshetnikov/ProveIt

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1 Results, comparison, and the exact logical assertion

1.1 What has been constructed

Fix the ordinary syntactic consistency statement

$$\mathrm{Con(ZFC)} \equiv \text{there is no finite derivation of falsity from the chosen ZFC axiom predicate.}$$

The repository identifies this predicate with $\neg$ `BProv ZFCax_s [] Form.fBot` at its formal endpoint. The constructions below are bound to the same literal 215-state machine and one complete native coefficient tuple.

Theorem 1.1 (Consistency equations, with stated inherited interfaces)**.** *Use the imported mathematical calibration of the literal blank-K215 compiler stated in Section 2, and the cited arithmetic interfaces, as theorems with ordinary proofs formalizable in ZFC. For every polynomial P constructed here, on its stated domain D, composition with the displayed reductions yields*

$$\mathrm{ZFC} \vdash (\neg\exists \boldsymbol{x} \in D\ P(\boldsymbol{x}) = 0) \longleftrightarrow \mathrm{Con(ZFC)}.$$

The eleven- and fourteen-positive-variable constructions use the ordinary Pell/compiler interface and do not use Sun's 2026 preprint. The nine-variable construction uses the repository's Jones interface; the ten-unrestricted-integer construction additionally uses Sun's ten-integer interface.

This is an ordinary provability claim obtained by composing the stated proof dependencies; assuming only their external truth would not suffice for the displayed internal conclusion. No consistency assumption is used. The new implications are proved below; the full unbounded compiler is inherited from the pinned repository, with its ordinary proofs included. We have not produced a new proof-assistant derivation of that compiler or an object-language ZFC proof term. Section 9 keeps these levels distinct.

The four principal outputs improve different measures. They cannot be ranked by one unqualified use of "simplest." In particular, the shortest known circuit in the reviewed family remains the historical 67-operation P_{67}, with fifteen positive variables and degree 155.

Domain / unknowns	Exact degree	Operations	Reviewed comparison
9 positive	30 947 584	651	Package 45: 129 875 980, 788
9 nonnegative	30 947 584	654	Package 45: same degree, 791
10 unrestricted	56 593 864	653	Package 40: 237 520 456, 782
11 positive	954	115	Package 38: 1140, 129; or 1154, 123
11 positive, system	2; 952	110	Two equations, shifted; equality tests free
14 positive	56	79	Historical Q_{56}: 15 variables, degree 56, 74 operations

The fourteen-variable lower-degree alternative and its distinct tradeoff are given in Section 8. The archive also retains an eleven-variable full-carrier variant of degree 958 at 116 operations, which provides a simpler comparison with package 44's twelve-variable carrier. The midpoint variant dominates it on the two displayed measures.

The new nine-variable degree is about 4.20 times smaller than the wave-6 comparison, and its evaluator removes 137 operations. The ten-integer degree is also about 4.20 times smaller and removes 129 operations from the degree-best comparison. Package 38 has a different ten-integer point at 773 operations and larger degree; the present 653-operation point improves that evaluator as well. The eleven-variable point removes 186 degrees and 14 operations from the previously lowest-degree eleven-variable equation, or 200 degrees and 8 operations from the 123-operation alternative.

1.2 Counting conventions and what is new

An operation is one binary addition, subtraction, or multiplication, including a multiplication by an integer numeral or a squaring. Shared registers are evaluated once. A fixed coefficient is a leaf, and its finite construction is accounted for separately. Fixed powers in a display are expanded into multiplication gates in the literal circuit. Thus these are circuit sizes, not counts of monomials in a fully expanded polynomial. All supplied variables, fixed ports, and emitted gates are live. For a system, comparisons are free under the same convention used by the repository. A natural variable includes zero; a positive variable does not.

The concrete contributions relative to the reviewed material are:

1. Only the 71 quadratic monomials that actually occur are isolated against all 528 possible quadratic expansion monomials. Full pair-sum uniqueness is unnecessary. This permits maximum weight 190 instead of 799.
2. Auxiliary-root recovery needs $d(Ig - cf) > b + cf$, rather than the stronger $Ig - cf > 1$. This removes a circular dependence on the index quotient being recovered.
3. The even polynomial modulus $U = 4q^4$ replaces a much larger carrier scale. Its gcd with H is at most three, and its congruence information is enough to recover an exact Pell-index equality.
4. The midpoint $\ell + k/2$ can be used in a carrier with integer coefficients because $U/2 = 2q^4$ is itself a polynomial. No additional variable or global doubling is needed.
5. An early index lower bound makes the same small modulus compatible with the ordinary strong-rank equation. This removes one variable while retaining degree 56, at the cost of five operations.

Sparse polynomial encodings, Pell equations, and relation-combining norms are established methods [2, 3, 4]. We claim the stated new combinations, finite weight assignment, explicit circuits, and comparisons with this repository revision. We do not claim the first use of support-sensitive Kronecker encoding in all of mathematics, an optimal weight assignment, or a global lower bound on the number of variables.

1.3 The reviewed baseline

The pinned commit is `124c07c0b77a9e20e84676514d8b1ff78536ea3f` [1], whose intake commit places wave-6 sources 38–45 beside sources 31–37. At that revision, the collection README still primarily summarizes wave 5. We therefore inspected the actual wave-6 source files and literal circuit data. The original incoming archives have been placed into the report directory; the pinned `docs/incoming` directory contains its placement ledger. The relevant current path is

`Computability/DiophantineConsistency/con-zfc-diophantine-forms/`.

Historical source paths in preserved files refer to the earlier `SetTheory/Cardinals` location at their own named commits. They are provenance, not instructions to mix old coefficient tuples.

2 One fixed coefficient tuple and the inherited interfaces

2.1 The native coefficients are fixed integers

Every new main construction uses the unchanged, complete package-31 tuple, whose base is

$$\rho = 2^{121}, \qquad L_c = 8801, \qquad B_c = \rho^{L_c} = 2^{1\,064\,921}.$$

The four ports are

$$\beta = B_c - 1, \qquad K_0 = D_C + B_c D_R, \qquad M_C, \qquad M_F = M_{F,\mathrm{native}} + B_c - 1.$$

The last port is already shifted. Substituting the unshifted high mask would define a different and unjustified equation.

The exact finite recipe and original table data are supplied in `coefficients/upstream/`; `coefficients/ports.json` binds the four materialized gzip payloads by full SHA-256 hashes and their original names. Their uncompressed sizes are

Port	Bit length	Bytes
β	1 064 921	133 116
K_0	1 070 004	133 751
M_C	1 064 921	133 116
M_F	1 064 922	133 116
Total		533 099

These integers do not depend on whether the machine halts. They are computed from its finite transition table and the fixed compiler layout. The nine- and ten-variable circuits additionally use fixed coefficient leaves obtained by explicit integer recipes from these four ports. Every such leaf has a name, recipe, and checked bound. None is an unspecified index supplied by an existence theorem.

We deliberately retain this common tuple to isolate the new arithmetic reductions. Package 44 supplies smaller native coefficients, but we do not claim a coefficient-size record or silently rebind an existing compression constant to that different tuple.

2.2 The semantic chain being imported

The inherited chain identifies three statements: a finite ZFC refutation exists; the literal blank K215 machine halts; and the calibrated positive compiler source has a solution. The machine endpoint is represented in the repository by the theorem

`KEndpoint215.blank_halts_iff_zfc_refutation`.

The ordinary mathematical compiler proofs, their source notices, and the selected coefficient adapter are preserved in this archive. The scalar Pell arguments are formalizable in ordinary arithmetic available inside ZFC; this observation is not an exported object-level proof term.

For the eleven-variable construction we use the following precise properties of the calibrated source:

1. From positive coordinates and the transport equation $N_t = 1$, the weak bounds in Lemma 5.1 hold.
2. The normalized Pell equations, including the reconstructed positive quotients h, γ, have signed-quotient soundness: a negative recovered strong root can be accompanied by the corresponding sign change of its quotient coordinate, and an accepting blank computation is still extracted.
3. A finite blank halt supplies canonical positive normalized witnesses with main Pell index R and first Pell index $(R+1)/2$.

These are the interfaces used by packages 38 and 44. We do not infer them from finite modular samples. For low-variable encoding, the two literal 22-equation quadratic parents are audited separately: package 40's parent and package 45's parent have the same monomial support but one different row after their affine coordinate choices. The archive checks the coordinate relation and does not replace one parent by the other without that transformation.

3 Support-aware quadratic packing

The main low-variable improvement is to replace full Sidon separation by separation only where the input system actually has a nonzero monomial. This changes the largest fixed exponent label from 799 to 190.

3.1 The exact hypothesis needed

Let $p_1, \dots, p_m \in \mathbb{Z}[X_1, \dots, X_n]$ have total degree at most two, let $\mathcal{A}$ be the union of their supports, and set $\mathcal{Q} = \{\alpha \in \mathbb{N}^n : |\alpha| \le 2\}$. For distinct positive integers w_i, write $e(\alpha) = \sum_i \alpha_i w_i$ and $W = \max_i w_i$. The sufficient condition is

$$\alpha \in \mathcal{A}, \quad \beta \in \mathcal{Q}, \quad e(\alpha) = e(\beta) \quad \Longrightarrow \quad \alpha = \beta. \tag{3.1}$$

The map e need not be injective on all of $\mathcal{Q}$. Let $h_j = 2W + (j-1)(2W+1)$, and define

$$c_{j,\alpha} = (2 - |\alpha|)! \prod_i \alpha_i! \, [X^\alpha] p_j,$$

$$D(T) = \sum_{j,\alpha} c_{j,\alpha} T^{h_j - e(\alpha)}, \qquad C_z(T) = \left(1 + \sum_i z_i T^{w_i}\right)^2.$$

Then all exponents of D are nonnegative and

$$[T^{h_j}] C_z(T) D(T) = 2 p_j(z) \qquad (1 \le j \le m). \tag{3.2}$$

Proof. A source monomial α of row k and an expansion monomial β can contribute to h_j only if $h_j - h_k = e(\beta) - e(\alpha)$. The right side has absolute value at most $2W$; a nonzero left side has absolute value at least $2W + 1$. Hence $k = j$, and (3.1) gives $\beta = \alpha$. The factorial in $c_{j,\alpha}$ cancels the multinomial denominator, giving (3.2). The row intervals $[h_j - 2W, h_j]$ are disjoint and the supported exponents within each are distinct. Therefore, if $|c_{j,\alpha}| < M$, each coefficient of D has absolute value less than M. Each coefficient of $C_z D$ has absolute value at most

$$M \left(1 + \sum_i |z_i|\right)^2. \tag{3.3}$$

Indeed, the sum of the nonnegative coefficients of $C_{|z|}$ is exactly the squared quantity. Collisions between absent monomials preserve this sum. In particular, no additional factor m is needed. □

3.2 The explicit instance

The variable order is the inherited order

$$J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma, N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B_{\mathrm{aux}}, \Delta.$$

In that order the weights are

$$(10, 14, 128, 112, 178, 157, 173, 145, 129, 63, 123, 84, 151, 75, 2, 124,$$
$$3, 169, 31, 190, 185, 184, 189, 177, 90, 111, 19, 30, 68, 51, 144).$$

With a padding weight zero, there are $\binom{33}{2} = 528$ degree-two multisets. The source uses only 71 of their monomials. Each supported sum is unique among all 528 sums, while the entire list has only 278 distinct values. Thus 250 repetitions are allowed, and the vector is explicitly not a B_2 set. The finite verifier reconstructs the source supports from the literal parent circuits and

checks all $136 \cdot 528 = 71{,}808$ source/expansion pairs. Its only 136 target hits are the required diagonal hits.

Two inherited parent presentations are retained exactly. The ten-integer construction uses package 40's 65-operation parent; the nine-variable construction uses package 45's 60-operation parent. Their supports agree, but their transport coordinates satisfy $z_{\mathrm{old}} = z_{\mathrm{new}} + 1$. The affine polynomial identity is checked independently. The inherited positive-zero correspondence, including the step $z_{\mathrm{old}} \geq 2$, is recorded in the bundled parent proof. Their coefficients are not silently identified. Both use the same complete four-port tuple of package 31, including the already shifted fourth port.

3.3 Why the unbounded source interface is preserved

First rewrite the inherited positive-variable parent by $x_i = z_i + 1$, so that the source coordinates z_i are nonnegative. Move its fifth shifted residual to the last row. Its value at the zero vector is 1, so the highest coefficient of D is 2, and the source has no common zero at the zero vector. Positivity of the weights and a geometric-series bound give $D(B) > 0$ whenever $B > M + 1$. All no-carry arguments use only this positive leading term, (3.2), (3.3), and the maximum weight. They do not need the full Sidon hypothesis.

For the nine-variable form use $h_j = 381j$, so $h_{22} = 8382$ and $K = h_{22} + 2W = 8762$. The unchanged bound on weighted coefficients is $M = 2^{2134927}$; put $b_0 = 4096M = 2^{2134939}$, $b = b_0 + \varepsilon$ and $B = b^3$. Define

$$\ell = \sum_i B^{w_i}, \qquad N_0 = B^{191}, \qquad N_1 = 4B^{8763}, \quad N = N_0 N_1,$$

$$M_0 = N_0 - 1 - (b-1)\ell, \qquad T_0 = \sum_j B^{h_j},$$

$$T = M_0 + (B-2)T_0 N_0,$$

$$J_0 = 2(1+g)^2 D(B) + B\sum_{k=0}^{8762} B^k, \qquad S = g + J_0 N_0,$$

$$R = (N-1)\big((S+T+1)N + T + 1\big), \quad G_{\mathrm{cap}} = 32bB^{190}.$$

The bounded branch $g < 4bB^{190}$ gives $g < N_0$ and $0 < J_0 < N_1$; also $0 < T < N$. When b is a power of two, the first mask forces $g = \sum_i z_i B^{w_i}$ with $0 \leq z_i < b$. Equation (3.3) gives strict coefficient magnitudes below $M(32b)^2 \leq B/4$. After adding $B/2$ to each coefficient, all digits lie strictly between zero and B. The second mask forces every target digit to equal $B/2$, and therefore every source row vanishes. The binary-popcount identity gives the inherited central-binomial divisibility condition $N^2 \mid \binom{2R}{R}$.

Conversely, choose an arbitrarily large power of two b above the coordinates of a common source zero. The resulting packed g is positive, all target tests pass, and $32g < G_{\mathrm{cap}}$. Thus both directions of the Jones ratio interface hold. For Sun's ten-integer interface use $h_j = 380 + 381(j-1)$, $K = 8761$, $b = 1 + 3f$ and $B = 2^{2140030}b^2$. The original coefficient majorant and all congruence conditions remain valid. The source cap is still bB^{190}, in precisely the required fixed-power form. The inherited finalizers therefore apply without increasing the number of witnesses.

3.4 Resulting fixed polynomials and certified metrics

The literal ten-integer circuit has 653 live binary arithmetic operations and exact total degree 56,593,864. The literal nine-positive-variable circuit has 651 operations and exact degree 30,947,584; its nonnegative-variable translation has 654 operations. For the latter, $t = \deg N =$

26,862, and the inherited cancellation-resolved identity gives

$$\deg F_9 = 1152t + 12W + 280 = 30{,}947{,}584,$$

with diagonal leading coefficient 2^{2050}. A separate certificate circuit has 666 operations but represents exactly the same polynomial; the 651-operation circuit is used for the operation comparison. All fixed powers in the delivered circuits are expanded into multiplication gates. Exact degree lower bounds follow from nonzero leading coefficients at the proved degree upper bound. Modular full-formula comparisons are reported separately as implementation evidence.

Against the reviewed wave-6 constructions, the degrees fall from 237,520,456 and 129,875,980, respectively. The operation savings are 129 and 137. The consistency equivalences retain their imported compiler/K215, Jones, and (for the integer form) Sun dependencies. This contribution proves and certifies the new source transformation; it does not claim global minimality or a new end-to-end proof-assistant certificate.

4 An explicit equation in eleven positive variables

The eleven supplied variables, and no others, are

$$J, F, \alpha, z, i, T, s, w, \eta, \zeta, t \in \mathbb{Z}_{>0}.$$

Use the four fixed ports above and define the polynomial abbreviations

$$\begin{aligned}
&q = \beta J + 1, && C = q - F - \alpha, && X = wq, && Y = sq^3,\\
&E = XY, && k = \eta + \zeta, && c = kY + \eta, && a = E + Y,\\
&H = 4a + 3, && \Delta = a^2 + H = (a+2)^2 - 1, && \ell = EkY, && A = 1 + \ell(\ell + k),\\
&G = 1 + \Delta c^2, && b_0 = X + ac, && I = ic^2, && D = 1 + \Delta I^2,\\
&Q = \Delta(D-1) = (\Delta I)^2, && d = Q - 1, && v = cT, && b = c + RD,
\end{aligned}$$

where the rank and transport polynomials are

$$R = ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J, \tag{4.1}$$

$$N_t = (K_0 + w)C + q - F - z(q-1). \tag{4.2}$$

Put

$$\begin{aligned}
&U = 4q^4, && r = k - R - 1, && K = IT,\\
&S = UH, && L = U + SK, &&\\
&W = U\big[b_0 + H(t + b + \ell)\big] + H\big(2q^4 k - r\big), &&&& \qquad (4.3)\\
&e = Q(Dv^2 + b^2) - 1, && B_a = 2Qvb. &&
\end{aligned}$$

The symbol K in (4.3) is a computed weight, distinct from the fixed coefficient K_0.

Define

$$\mathcal{N} = \prod_{\substack{g=\pm\sqrt{G},\ \tau=\pm\sqrt{A}\\ f=\pm\sqrt{D}}} \Big[d(W - Lg - S\tau)^2 - S^2 e + S^2 B_a f\Big]. \tag{4.4}$$

This is an integer polynomial: the product is invariant under each formal sign change, and only even powers of the three formal roots remain. This definition does not assume that the square classes are independent or that there exist eight independent field automorphisms. The radical symbols are notation for a finite polynomial identity.

Our main equation is

$$\boxed{P_{11} = N_t(1+\mathcal{N}) - 1 = 0.} \tag{4.5}$$

Every named quantity in (4.1)–(4.5) is computed from the eleven supplied variables. There is no variable exponent, divisibility predicate, square root operation, or extra existential quantifier in the final literal circuit.

4.1 A division-free scalar formula

The following evaluation specifies (4.4) using only integer additions, subtractions, and multiplications:

$$\begin{aligned}
n_d &= dS^2A, & P_H &= S^2QDv^2, & U_H &= S^2Qb^2,\\
e_H &= P_H + U_H - S^2, & j &= 4P_HU_H,\\
\mathcal{T} &= 2d(W^2 + L^2G), & \kappa &= [d(W^2 - L^2G)]^2,\\
\mathcal{A} &= 2(n_d + e_H), & \mathcal{B} &= (n_d - e_H)^2 - j,\\
a_r &= \mathcal{T} - \mathcal{A}, & b_r &= \mathcal{B} - \kappa,\\
v_r &= 16n_dj - a_r(a_r\mathcal{T} + 2b_r), & u_r &= b_r^2 - a_r^2\kappa,\\
\mathcal{N} &= \kappa v_r^2 - u_r(\mathcal{T}v_r - u_r).
\end{aligned}$$

For a proof, after eliminating the signs of τ and f, the polynomial in $x = d(W - Lg)^2$ is

$$(x^2 - \mathcal{A}x + \mathcal{B})^2 - 16n_djx.$$

The two values obtained from the signs of g have sum $\mathcal{T}$ and product κ. Reduction modulo $x^2 - \mathcal{T}x + \kappa$ gives $u_r - v_rx$; its quadratic norm is the displayed expression. This proves the scalar identity over an integer polynomial ring, including all square and zero specializations.

Lemma 4.1 (The unit guard)**.** *For every integer substitution, $\mathcal{N}$ is congruent to a square modulo four. Hence a root of (4.5) satisfies $N_t = 1$ and $\mathcal{N} = 0$.*

Proof. Set $x_0 = d(W^2 + L^2G)$. Then $\mathcal{T} = 2x_0$ and $\kappa = x_0^2 - 4d^2W^2L^2G$. Thus

$$\mathcal{N} \equiv (u_r - x_0v_r)^2 \pmod 4.$$

At a product root, N_t and $1 + \mathcal{N}$ are integer units. The possibility $1 + \mathcal{N} = -1$ would require $\mathcal{N} = -2$, impossible modulo four. Both units are therefore $+1$. □

5 Root recovery without assuming the missing quotient

Lemma 5.1 (Weak bounds)**.** *At a positive root of (4.5), the authentic coefficient interface and (4.2) give*

$$\begin{gathered}
q \ge 16, \quad C > 0, \quad 3q < R < q^4 - 2, \quad X \ge q, \quad Y \ge q^3, \quad E \ge q^4 > R + 2,\\
k \ge 2, \quad kY < c < k(Y+1), \quad I \ge c^2, \quad a + 2 < 2E.
\end{gathered}$$

In particular $c > 8192 > 96$. No inequality $k > E$ is used here.

Proof. The transport and rank bounds are the imported weak interface. In particular, $C \le 0$ would make $N_t \le 1 - F - (z-1)(q-1) \le 0$, contradicting $N_t = 1$. All remaining assertions follow from the positive coordinate definitions; $a + 2 = E + Y + 2 < 2E$ uses $X \ge 16$ and $Y \ge 4096$. □

In the following estimates, write $g = \sqrt{G}$, $f = \sqrt{D}$, and $\tau = \sqrt{A}$ for the positive real roots, whether or not they are integers. The elementary inequalities

$$1 < A < \Delta I = \sqrt{Q}, \qquad g < (a+2)c < 2Ec, \qquad f < (a+2)I < 2EI \tag{5.1}$$

hold. For the first, $c > kY$ gives

$$\Delta c^2 - A > Ek^2Y(2Y^2 - 1) - 1 > 0,$$

and $I \geq c^2$. Notice that these estimates do not use an already recovered main or first Pell index.

Lemma 5.2 (The refined auxiliary gap)**.** *Under the weak bounds,*

$$d(Ig - cf) > b + cf.$$

Every coherent complex auxiliary root satisfying

$$dy^2 = Q(vf' - b)^2 - 1, \qquad (f')^2 = D,$$

therefore satisfies $|y| < b + ITg$.

Proof. The exact relation

$$(Ig)^2 - (cf)^2 = I^2 - c^2$$

and $I \geq c^2$, $c \geq 2$, give

$$Ig - cf = \frac{I^2 - c^2}{Ig + cf} > \frac{I^2}{4Ig} > \frac{I}{8Ec}.$$

Also $d > \frac{1}{2}E^4I^2$. Since $R < E$, $D < 4E^2I^2$, and $c < I^2$, one has $b + cf < 6E^3I^2$. Consequently

$$d(Ig - cf) > \frac{E^3I^3}{16c} > 6E^3I^2 > b + cf,$$

where the middle strict inequality uses $I > 96c$.

Set $Z = b + Tcf > 1$. Even for a complex y, its defining equation implies $d|y|^2 \leq QZ^2 + 1$. The exact difference

$$[Z(1 + 1/d)]^2 - \frac{QZ^2 + 1}{d} = \frac{QZ^2 - d}{d^2} > 0$$

gives $|y| < Z + Z/d$. As $T \geq 1$, $Z/d \leq T(b + cf)/d < T(Ig - cf)$. Thus $|y| < b + ITg$. □

This is the compatibility lemma missing from a direct juxtaposition of packages 38 and 44: the latter's estimate $Ig - cf > 1$ used the old index quotient. The refined inequality is valid even when $k \leq E$.

5.1 The midpoint and selection of the main root

For the proof only, put $\delta = \ell + k/2$. The carrier (4.3) has integer coefficients even if this temporary rational abbreviation is not yet an integer. Its inner argument is

$$x = \frac{W - Lg'}{S} = t + b + \delta - \frac{r}{U} + \frac{b_0 - g'}{H} - ITg', \qquad (g')^2 = G. \tag{5.2}$$

Since $k \geq 2$,

$$\delta \geq \tau, \qquad 0 \leq \delta - \tau = \frac{k^2/4 - 1}{\delta + \tau} < \frac{k}{8EY} < Ig - cf. \tag{5.3}$$

The last inequality follows from Lemma 5.2's stronger bound $Ig - cf > I/(8Ec) \geq c/(8E) > kY/(8E)$. We do *not* claim $\delta - \tau < 1$; k is unbounded.

Each root of the inner elimination polynomial has a coherent form $x = \tau' + y$ with $(\tau')^2 = A$ and the auxiliary relation in Lemma 5.2. Thus $|x| < \tau + b + ITg$. If $g' = -g$, however, (5.2) gives

$$x = t + b + \delta - r/U + (b_0 + g)/H + ITg > b + \tau + ITg.$$

Indeed $H < 5a$, $b_0 > ac$, $g > ac$, and $c > kY$ imply $(b_0 + g)/H > 2c/5 > k > r/U$. This is a contradiction. If G were nonsquare, actual conjugation of $\mathbb{Q}(\sqrt{G})$ would carry the vanishing rational-coefficient inner polynomial at the positive branch to the forbidden negative branch. Therefore g is a positive integer and x is rational. This argument uses one actual quadratic conjugation, not fictitious independent signs in a splitting field.

5.2 Exceptional inner roots and rational Pell saturation

The inner polynomial is

$$C_x = d(x^2 + A) - e,$$
$$F_*(x) = \left[C_x^2 + DB_a^2 - 4d^2x^2A\right]^2 - 4C_x^2DB_a^2.$$

It has $F_*(x) = 0$ for the rational x just recovered. If $C_x = 0$, substitution gives the integer identity

$$d^2A^2 - deA + Q^2v^2b^2D = 0.$$

Reducing modulo Q, where $d \equiv e \equiv -1$, gives $Q \mid A(A-1)$, impossible since $0 < A(A-1) < Q$. Likewise $F_*(0) = 0$ would give $Q \mid (A-1)^2$, also impossible. Therefore both x and C_x are nonzero. Now define

$$U_x = C_x^2 + DB_a^2 - 4d^2x^2A,$$
$$f' = -\frac{U_x}{2C_xB_a}, \qquad \tau' = \frac{C_x + B_af'}{2dx}, \qquad y = x - \tau'.$$

All denominators have been justified. Direct substitution gives $(f')^2 = D$, $(\tau')^2 = A$, and

$$dy^2 = Q(vf' - b)^2 - 1.$$

The rational square roots f', τ' of integers are integers.

We also need $y \in \mathbb{Z}$. The inherited rational Pell saturation lemma states that, for integers $m \geq 2$ and V, if

$$m^2V^2 - (m^2 - 1)y^2 = 1, \qquad y \in \mathbb{Q},$$

then $y \in \mathbb{Z}$. Here $m = \Delta I$ and $V = vf' - b$. For completeness, write $y = p/q$ in lowest terms. Then $q^2 \mid m^2 - 1$, and the positive nonsquare $D_0 = (m^2 - 1)/q^2$ has integer-coordinate Pell solutions (m, q) and $(m|V|, |p|)$. Both are powers of its fundamental integer-coordinate unit. Their abscissas χ_j, χ_l satisfy $\chi_j \mid \chi_l$. Reducing and reflecting l modulo $2j$, the addition formulas and strict growth show that l is an odd multiple of j. The ordinate formulas then give $q \mid p$; lowest terms force $q = 1$. The cases $V = 0$ or $p = 0$ cannot satisfy the original equation. It follows that y, and hence x, is integral.

6 A small modulus recovers the deleted Pell quotient

Write $A_0 = a + 2$ and $P_0 = 2EY + 1$. The first Pell equation has fundamental integer-coordinate unit $(P_0, 2)$, while the main equation has fundamental unit $(A_0, 1)$. Thus

$$|\tau'| = \chi_{P_0}(n), \quad k = 2\psi_{P_0}(n), \qquad g = \chi_{A_0}(p), \quad c = \psi_{A_0}(p) \tag{6.1}$$

for positive indices n, p. In particular k is even. Only now do we assert that δ is integral. From the integer carrier identity (5.2), it follows that

$$U \mid Hr. \tag{6.2}$$

Cancelling H at this stage would be invalid.

6.1 Enough index separation without translating s

Because $c > 1$, we have $p \geq 2$. If $n = 1$, then $k = 2$ and positivity forces $\eta = \zeta = 1$, so $c = 2Y + 1$. But $c = \psi_{A_0}(p) \geq 2A_0 > 2Y + 1$, a contradiction. Hence

$$n \geq 2, \qquad k \geq 4P_0 > 8EY, \qquad c > 8EY^2 > 2R, \qquad c > X. \tag{6.3}$$

The standard growth estimate also gives $c > 2p$. Thus the necessary index separation comes from the integral Pell equations themselves. No replacement $s_{\mathrm{old}} = s + 3q$ is necessary.

The strong equation gives

$$|f'| = \chi_{A_0}(m), \qquad I = \psi_{A_0}(m)$$

for $m \geq 1$. Strong divisibility and $c^2 \mid I$ first give $p \mid m$. If $m = pj$, composition yields

$$\psi_{A_0}(m) = c\psi_{\chi_{A_0}(p)}(j).$$

The second factor is congruent modulo c to $j\chi_{A_0}(p)^{j-1}$. Since $\chi_{A_0}(p)$ is a unit modulo c, we obtain $c \mid j$, hence $pc \mid m$ and $m > 2p$.

The auxiliary equation is

$$(\Delta I)^2 V^2 - ((\Delta I)^2 - 1)y^2 = 1, \qquad V = c(Tf' - 1) - R(f')^2.$$

Its Pell index j_0 is positive and odd. The usual odd-index abscissa polynomials Q_v satisfy

$$\chi_M(2v + 1) = MQ_v(M^2),$$
$$Q_v(1 - A_0^2) = (-1)^v \psi_{A_0}(2v + 1), \qquad Q_v(0) = (-1)^v(2v + 1).$$

Reducing modulo $|f'|$, using $Q = \Delta((f')^2 - 1) \equiv 1 - A_0^2$ and $V \equiv -c$, gives $\psi_{A_0}(j_0) \equiv \pm c$. Squaring and doubling gives

$$\chi_{A_0}(2j_0) \equiv \chi_{A_0}(2p) \pmod{\chi_{A_0}(m)}.$$

Since $0 < 2p < m$, signed index reduction gives $j_0 \equiv \pm p \pmod{m}$. To see why the signed step is valid, reduce an index modulo $2m$ and reflect it to $[0, m]$. A negative representative cannot equal $\chi_{A_0}(2p)$ modulo $\chi_{A_0}(m)$ because $2\chi_{A_0}(m - 1) < \chi_{A_0}(m)$; strict growth then forces the representative to be $2p$.

Separately, $Q \equiv 0 \pmod{c}$ and $V \equiv -R \pmod{c}$ give $j_0 \equiv \pm R \pmod{c}$. Since $c \mid m$ and $0 < p, R < c/2$, these congruences force

$$p = R. \tag{6.4}$$

No step used an index quotient before its reconstruction.

Lemma 6.1 (The modulus $4q^4$)**.** *Under the preceding hypotheses,* $2n = R + 1$*, and both* E *and* U *divide* $r = k - R - 1$*.*

Proof. The polynomial $H = 4sq^3(wq+1)+3$ is odd. Any prime dividing both q and H must be three. If $3 \mid q$, then $H \equiv 3 \pmod{27}$, so its 3-adic valuation is exactly one. Consequently

$$\gcd(U,H) \in \{1,3\}, \qquad \mu = U/\gcd(U,H) \geq 4q^4/3 > R.$$

The Pell recurrence gives $\psi_{P_0}(n) \equiv n \pmod{P_0-1}$, hence

$$k \equiv 2n \pmod{4EY}.$$

Moreover $U = 4q^4 \mid 4EY = 4ws^2q^7$. Combining this with (6.2) gives $2n \equiv R+1 \pmod{\mu}$. Since $P_0 > A_0$ and $c > k$, monotonicity of the Pell sequences in their base and index implies $n < p = R$. Thus $|2n-R-1| < R < \mu$, and the congruence is an equality. It follows that $r = k-2n$ is divisible by $4EY$, hence by E and U. □

Set

$$h = r/E, \qquad \gamma = (g-b_0)/H.$$

The first is a positive integer because $n \geq 2$ and $\psi_{P_0}(n) > n$. The carrier, $U \mid r$, and even k give $g \equiv b_0 \pmod{H}$, so the second is an integer. Since $g > (a+1)c$ and $c > X$, it is positive. We have now reconstructed the normalized source equations

$$(\tau')^2 - \ell(\ell+k) = 1, \qquad (b_0+H\gamma)^2 - \Delta c^2 = 1,$$
$$(f')^2 - \Delta i^2 c^4 = 1, \qquad Q[c(Tf'-1) - R(f')^2]^2 - (Q-1)y^2 = 1,$$
$$k - hE - R = 1, \qquad N_t = 1.$$

Take positive absolute values of the first and auxiliary roots. If $f' < 0$, change f' and its quotient coordinate T together, as allowed by the inherited signed-quotient soundness interface. This extracts a finite blank-K215 halt and hence a ZFC refutation.

7 Constructive completeness and exact size of the eleven-variable equation

7.1 Constructing the new positive coordinate

Conversely, take a canonical positive normalized compiler witness. It has positive g, τ, f, y, γ and first/main indices satisfying $R = 2n-1$. The Pell recurrence gives $r = k-2n \equiv 0 \pmod{4EY}$, so r/U is a positive integer; k is even as well. Define

$$t = \gamma + ITg + \tau + y - b - \delta + r/U, \qquad \delta = \ell + k/2. \tag{7.1}$$

The auxiliary Pell equation gives $y \geq |vf - b| \geq b - vf$. By (5.3),

$$t \geq \gamma + T(Ig - cf) + (\tau - \delta) + r/U > \gamma + r/U > 0.$$

Thus (7.1) is an integer positive coordinate. Substituting it into the carrier makes one coherent factor of (4.4) vanish, while $N_t = 1$ is unchanged. This proves the converse. Together with soundness, it establishes the eleven-variable case of Theorem 1.1 through the stated inherited calibration.

7.2 Exact degree, including the cancellations

A naive gate-degree propagation gives 1138 for the product. The actual degree is 954 because of exact norm cancellations. The following calculation proves the smaller degree, rather than inferring it from random evaluations.

Use a subscript zero for a highest homogeneous part, and set

$$q_0 = \beta J, \quad k_0 = \eta + \zeta, \quad X_0 = wq_0, \quad Y_0 = sq_0^3, \quad E_0 = X_0Y_0, \quad c_0 = k_0Y_0,$$
$$I_0 = ic_0^2, \quad H_0 = 4E_0, \quad \ell_0 = E_0c_0, \quad d_0 = E_0^4I_0^2,$$
$$R_0 = q_0^3(q_0 - F), \quad B_0 = R_0E_0^2I_0^2, \quad V_0 = I_0T\ell_0, \quad U_0 = 4q_0^4.$$

Their relevant degrees are $\deg H_0 = 6$, $\deg \ell_0 = 11$, $\deg d_0 = 46$, $\deg B_0 = 38$, and $\deg V_0 = 23$.

First omit the factor U and the rational shift $-r/U$ from the carrier, retaining the rational midpoint $\ell + k/2$. Work in the Laurent series field obtained by scaling all supplied variables by one indeterminate; its coefficient field is the rational function field in those variables and the fixed ports. The principal roots have leading terms $g_0 = \tau_0 = \ell_0$ and $f_0 = E_0I_0$. The exact square-difference identity gives

$$T(Ig - cf) = \frac{iTc_0}{2E_0} + \text{terms of smaller degree.}$$

For signs (s_g, s_τ, s_f), four factors with $s_f \neq s_g$ have degree 119 and leading term $-s_g 4H_0^2 d_0 B_0 V_0$. For the four factors with $s_f = s_g$, the leading terms are

s_g	s_τ	Degree	Leading term
$+$	$-$	107	$4H_0^2B_0d_0\ell_0$
$-$	$-$	107	$4H_0^2B_0d_0\ell_0$
$-$	$+$	101	$4H_0B_0d_0\ell_0$
$+$	$+$	97	$H_0^2B_0d_0(2t - iTc_0/E_0)$

To verify the table, define the full unscaled carrier expressions

$$W^{\mathrm{u}} = b_0 + H(t + b + \ell + k/2), \qquad L^{\mathrm{u}} = 1 + HIT,$$

and put $Z = W^{\mathrm{u}} - Hb - Hs_\tau\tau - L^{\mathrm{u}}s_g g$. Expand each factor as

$$2dHbZ + dZ^2 - H^2b^2 - H^2QDv^2 + H^2 + s_f 2H^2Qbvf.$$

The leading degree-119 terms cancel exactly when $s_f = s_g$. In that case the residual linear expression is controlled successively by 2ℓ, $(b_0 + g)/H \sim 2\ell_0/H_0$, or $t - T(Ig - cf)$; the midpoint error $\delta - \tau$ has degree -9. The quadratic cancellation follows from

$$dI^2G - QDc^2 = Q(I^2 - c^2) - I^2G,$$

so the remaining quadratic terms are below the listed degrees. This also supplies a uniform upper bound, not just a leading-term guess.

Multiplying the table gives unscaled norm degree $4 \cdot 119 + 107 + 107 + 101 + 97 = 888$ and leading form

$$2^{14}H_0^{15}d_0^8B_0^8V_0^4\ell_0^3\left(2t - \frac{iTc_0}{E_0}\right).$$

The actual norm is, in the localization at U,

$$\mathcal{N} = U^{16}\mathcal{N}_{\mathrm{unscaled}}(t - r/U).$$

Its degree in the carrier coordinate is at most sixteen, so the denominators clear. Since $\deg(r/U) = 0 < \deg t$, the substitution does not alter the highest homogeneous part. Consequently

$$\mathcal{N}_{952} = 2^{14} U_0^{16} H_0^{15} d_0^8 B_0^8 V_0^4 \ell_0^3 \left(2t - \frac{iTc_0}{E_0}\right) \neq 0. \tag{7.2}$$

The denominators in the displayed expression cancel against its prefactors. Nonvanishing follows from the independent occurrence of t, and holds for every authentic $\beta > 0$. Finally

$$(N_t)_2 = w(q_0 - F - \alpha) - zq_0 \neq 0,$$

so (4.5) has exact degree $952 + 2 = 954$.

7.3 Operations and an optional system

The literal schedule has 69 multiplications and 46 additions or subtractions, totaling 115. The use of $L = U + SIT$ shares the already computed S, and the midpoint term uses $U/2 = 2q^4$ directly. The exact two-equation form $N_t = 1$, $\mathcal{N} = 0$ uses 111 operations if the final norm is tested by comparing its two terms. Applying the inherited positive transport shift

$$(K_0 + w)C = F + z(q-1), \qquad z_{\text{old}} = z + 1,$$

gives the 110-operation system of degrees 2 and 952. Canonical witnesses have $w = 2^R/q > q > F$, so $z_{\text{old}} \geq 2$; the shifted coordinate is positive. Arbitrary positive shifted solutions map back by $z_{\text{old}} = z + 1$.

The checker computes complete univariate polynomials modulo 10007 and 10009 using the actual materialized coefficients. It obtains nonzero degree-954 coefficients 8551 and 2035 on the two recorded rays, matching (7.2). These are exact finite-field calculations, not interpolation. The paper supplies the upper bound; the nonzero coefficient supplies an independent lower-bound certificate for the literal instance.

8 A fourteen-coordinate equation of degree 56

The quotient absorber also applies before radical elimination. Combined with the inherited ordinary strong norm, it gives fourteen strictly positive unknowns, exact degree 56, and 79 binary arithmetic operations. The historical Q_{56} has fifteen positive unknowns, degree 56, and 74 operations. Thus the construction exchanges five operations for one fewer unknown, while retaining the same degree. This is a separate tradeoff from the equations in eleven or fewer coordinates.

Keep the common definitions of $q, C, X, Y, E, k, c, a, H, \Delta, R, b_0$, with $Y = sq^3$ and $b_0 = X + ac$. The supplied coordinates are

$$J, F, \alpha, z, f, i, T, s, w, \tau, \eta, \zeta, y, \gamma > 0.$$

In addition to $U = 4q^4$ and $r = k - R - 1$, put

$$\begin{aligned} V &= c(Tf - 1) - Rf^2, & G &= 1 + \Delta c^2, \\ N_0 &= \tau^2 - (XY^2)(XY^2 + 1)k^2, & I &= ic^2, \\ A_{\text{aux}} &= \Delta(f^2 - 1), & N_a &= A_{\text{aux}}(V^2 - y^2) + y^2, \\ N_t &= (K_0 + w)C + q - F - z(q-1), & D_* &= Ub_0 + H(U\gamma - r). \end{aligned}$$

The single ordinary integer polynomial equation is

$$\boxed{(D_*^2 - U^2 G)^2 + (I^2 - A_{\text{aux}})^2 + (N_a - 1)^2 + (N_0 N_t - 1)^2 = 0.} \tag{8.1}$$

Every abbreviation is computed, not another unknown. The same complete package-31 coefficient tuple supplies all four fixed ports. The coordinate i now has the ordinary strong meaning: the equation is $I^2 = \Delta(f^2 - 1)$. One must use the ordinary strong-rank theorem instead of canceling Δ informally.

An early lower bound on the index. At a zero of (8.1), all four residuals vanish. The consecutive-product negative-Pell descent excludes $N_0 = -1$, so $N_0 N_t = 1$ gives $N_0 = N_t = 1$. The authentic transport interface yields $q \geq 16$, $3q < R < q^4 - 2$, and $E \geq q^4 > R + 2$. The main residual makes $g = D_*/U$ a rational square root of the integer G, hence an integer. It is positive: $H < 5a$, $b_0 > akY$, and $UY > 5$ give $D_* > Ub_0 - Hk > ak(UY - 5) > 0$. The carrier implies $U \mid Hr$.

Write $A_* = a + 2$ and $P_0 = 2XY^2 + 1$. Integer-coordinate Pell classification supplies positive indices p, n with

$$c = \psi_{A_*}(p), \quad g = \chi_{A_*}(p), \quad k = 2\psi_{P_0}(n).$$

Because $P_0 > A_*$ and $c > k$, we have $p > n$. The recurrence gives $k \equiv 2n \pmod{4XY^2}$, and $U \mid 4XY^2$. Also $\gcd(U, H) \leq 3$: H is odd, every common prime divisor of q and H divides 3, and $3 \mid q$ implies $H \equiv 3 \pmod{27}$. Therefore

$$\mu = U/\gcd(U, H) \geq 4q^4/3 > R + 1.$$

The divisibility $U \mid H(k - R - 1)$ gives $2n \equiv R + 1 \pmod{\mu}$. Since $n > 0$ and $0 < R + 1 < \mu$,

$$2n = R + 1 + j\mu \quad (j \geq 0).$$

Thus $n \geq 25$ and $p \geq 26$ *before* recovering $p = R$. The elementary recurrence bound $c \geq (2A_* - 1)^{p-1} \geq A_*^{25}$ gives

$$c > A_*\Delta^2, \qquad c > 2p, \qquad c > 2R.$$

For the last inequality one may also use $c \geq 2A_* > 2R$.

Ordinary strong-rank recovery. Apply the inherited ordinary strong-rank lemma, with its strict size hypothesis now proved. From $c = \psi_{A_*}(p) > A_*\Delta^2$ and $(ic^2)^2 = \Delta(f^2 - 1)$ it gives a positive index m such that

$$f = \chi_{A_*}(m), \quad p \mid m, \quad c \mid m, \quad m \geq c > 2p, \quad I = \Delta\psi_{A_*}(m), \quad c^2 \mid f^2 - 1.$$

The proof, reproduced among the inherited sources, handles the possible nonfundamentality of A_* in the squarefree Pell group. Its large-c condition is essential to this invocation. No assertion that I/Δ is integral follows merely from the original displayed strong equation.

Set $Q = I^2 = A_{\mathrm{aux}}$. The auxiliary equation becomes $QV^2 - (Q - 1)y^2 = 1$. Its odd Pell index j_0 satisfies $j_0 \equiv \pm p \pmod{m}$ by reduction modulo f and signed abscissa reduction; it satisfies $j_0 \equiv \pm R \pmod{c}$ by reduction modulo c. The latter uses the recovered divisibility $c^2 \mid f^2 - 1$. Since $c \mid m$ and $0 < p, R < c/2$, it follows that $p = R$. These are exactly the ordinary signed auxiliary steps in the inherited compiler proof, with all required bounds available.

Now $1 \leq n < p = R$, so $|2n - R - 1| < R < \mu$. The earlier congruence forces $j = 0$ and $2n = R + 1$. Hence $r = k - 2n$ is divisible by both E and U. Define

$$h = r/E, \qquad \gamma_{\mathrm{old}} = \gamma - r/U = (g - b_0)/H.$$

Both are integers. The first is positive since $n \geq 25$ and $\psi_{P_0}(n) > n$. Also $k \geq 4P_0 > X$ gives $c > X$, while $g > (a + 1)c$ implies $g - b_0 > c - X > 0$, proving $\gamma_{\mathrm{old}} > 0$. All fifteen positive coordinates of the ordinary-strong blank compiler have been restored, so its calibrated soundness theorem supplies the ZFC refutation.

Constructive converse. A positive ordinary-strong compiler witness satisfies $R = 2n - 1$. Thus $r = k - 2n$ is a positive multiple of $4XY^2$, hence of U. Set $\gamma = \gamma_{\mathrm{old}} + r/U$ and retain the other thirteen coordinates. Then $D_* = Ug$, and all four residuals in (8.1) vanish. No new condition on s or increase of a Pell index is needed. Canonical normalized witnesses give ordinary witnesses by the established substitution $i_{\mathrm{ordinary}} = \Delta i_{\mathrm{normalized}}$. This proves equivalent positive solvability and hence the stated equivalence of unsolvability with Con(ZFC).

Exact degree and reproducibility. The four residual degrees are $26, 22, 28, 24$. For the first use

$$D_*^2 - U^2 G = U^2((b_0 + H\gamma)^2 - G) - 2UHr(b_0 + H\gamma) + H^2 r^2.$$

The three terms have degrees 26, at most 25, and at most 20; the ordinary main residual in parentheses has exact degree 18. Only $(N_a - 1)^2$ reaches degree 56, with highest homogeneous form

$$T^4 f^8 w^4 s^8 (\eta + \zeta)^4 (\beta J)^{28}.$$

Its $J^{28}\eta^4 T^4 f^8 w^4 s^8$ coefficient is $\beta^{28} \neq 0$. The independent standard-library checker expands the complete main residual into 2,994 monomials with the coefficient ports symbolic, proves its degree 26, and checks all 79 live gates and the final nonzero highest form. This finite certificate verifies the expression and degree. The unbounded semantic dependencies remain the inherited ordinary compiler and machine calibration.

9 Verification, integration, and what remains unformalized

9.1 The checks and their proper interpretation

The archive separates finite implementation checks from mathematical soundness. Its main checks are:

- symbolic reconstruction of both quadratic parents and all 136 source coefficient records, with a separate check of their affine transport-coordinate relation;
- exhaustive isolation of 71 supported monomials against all 528 quadratic pairs, and all $136 \cdot 528 = 71\,808$ source/expansion tests at the target positions;
- exact fixed-leaf recipes, coefficient bounds, byte hashes, operation counts, variable counts, and complete circuit liveness;
- exact norm identities proved algebraically, with independent formal-radical multiplication diagnostics including square, zero, and coincident-radicand specializations;
- exact degree certificates and tests using the actual fixed tuple;
- direct accepting and rejecting finite controls for the source packing, where supplied, and the positive coordinate transformations.

None of these finite tests attempts to verify that an unbounded machine never halts. No search for a ZFC contradiction or for a gigantic zero was performed. An exhaustive combinatorial isolation certificate is different from a finite sample of an infinite semantic implication; the former is precisely the finite hypothesis of the encoding theorem.

9.2 Formalization status

At the pinned revision the active research records a Lean-checked nine-natural-variable existence theorem with coefficients selected through existential representation, and a checked quartic alternative. It explicitly leaves executable coefficient export, the explicit P_{67} compiler correctness in Lean, and an object-level ZFC derivation open. Those facts are preserved in the provenance snapshots.

The present article supplies explicit data and ordinary proofs for the new adapters. It does not silently turn the existing existence-based Lean polynomial into the literal circuits here, or claim that local ring identities prove the compiler endpoint. The new degree and operation certificates are finite certificates for specific polynomials; the intended consistency interpretation still composes with the inherited semantic chain described earlier. The mathematical reviews in `review/` were produced as independent AI-assisted reviews within this work, not as external human peer review.

9.3 Reproducing and incorporating the package

Run `python3 verify.py` from the archive root. Python 3.10 or later and NumPy are sufficient; the support-encoding and fourteen-variable checks themselves use the standard library. The verification driver regenerates the literal circuit files and writes an ordered receipt. The supplied `Makefile` rebuilds this article with a standard `pdflatex` installation. No network or running ProveIt checkout is needed for these finite checks.

The intended integration unit is the complete archive directory. Preserve the coherent coefficient tuple, source ledgers, and licenses. No existing repository files or remote branches were changed in this work. A suitable destination is a new descriptively named source directory under the current DiophantineConsistency report collection; its number should be assigned at intake instead of guessed here. The source commit is pinned so that later repository edits do not alter the comparison silently.

10 Further research questions

1. **How small can the support-dependent weights be?** The weight-190 certificate is a feasible construction, not an optimum. A reproducible SAT or integer-programming search could prove lower bounds for this exact 71-monomial support. That would distinguish a search improvement from a structural lower bound.
2. **Can row targets overlap?** The present theorem retains separated row blocks. The true obstruction is a finite set of support/expansion differences. Carefully certified nonuniform target placement may reduce the maximum target without requiring a globally stronger pair-sum condition.
3. **Can the quadratic parent be redesigned for its support?** Affine coordinate changes and choices of intermediate multiplication registers change which monomials need isolation. Jointly minimizing parent support and encoding range may outperform optimizing weights after the parent has been fixed.
4. **Can more coefficient or guard structure be retained?** The present carry estimate uses a safe total coefficient bound. Row-specific signs, shared coefficient expressions, and smaller independent buffers may reduce the fixed source base or evaluator, but each change must preserve the malformed-digit rejection argument.
5. **Can the new quotient lemma be formalized locally?** The modulus $4q^4$, its gcd bound, the Pell recurrence congruence, and the short-index argument are concrete targets for Lean. Their proof should explicitly order root recovery, evenness of k, weak divisibility, and restoration of h.
6. **Can the full semantic bridge be closed?** The most valuable formalization work is a data-valued compiler that connects the actual four ports and literal source circuit to the existing K215 endpoint. A local simplification certificate alone does not settle that obligation.
7. **Can the midpoint carrier remove another coordinate?** The refined root-gap estimate uses actual coupling between Pell radicands. A further removal must preserve rational-root saturation, the nonzero exceptional denominators, and positivity in completeness. No sub-eleven moderate-degree construction is asserted here.

A A finite native coefficient recipe

This appendix records the selected tuple without an unspecified enumeration index. The exact ordered tile list consists of the 943 quadruples in `coefficients/upstream/upstream/wang943_tiles.json`; the accompanying generator reconstructs it from the 430 literal K215 actions. Slots are west, north, east, south; start is selector zero, the impossible tile is selector one, and the remaining tiles are lexicographically ordered, as in package 31.

Let

$$\begin{aligned}\mathcal{E}_c = {} & \{0,\dots,942\}\cup\{983,\dots,992\}\cup\{1003,\dots,1032\}\\ & \cup\{1063,1064,1095,1096,2195,3296\}.\end{aligned}$$

It has 989 elements. Set $W_j = 2^{10+2j}$ for $0 \le j < 44$, and copy offsets $o_t = (1023, 1013, 1003, 983)$. If tile i has slot labels x_{it}, define

$$c_i = 1 + \sum_{t=0}^{3}\sum_{j=0}^{9} \mathrm{bit}_j(x_{it}+512)W_{10t+j} + W_{40} + W_{41} + W_{42} + W_{43}.$$

At exponent $o_t + j$ put $c_e = W_{10t+j}$; at the four listed anchors put $W_{40},\dots,W_{43}$. The two dummy coefficients at 1063 and 1064 are zero. Define

$$\begin{aligned}\mu_c &= 1022 + \sum_{j=0}^{43} W_j,\\ Z_c &= (2^{119}-1)\sum_{e=8792}^{8798}\rho^e + (2^{27}-1)\rho^{8799},\\ D_C &= \rho^{30}+\rho^{62}+\rho^{1100}+\rho^{2200}+\sum_{e\in\mathcal{E}_c} c_e(\rho^{4393-e}+\rho^{5495-e}),\\ D_R &= \rho^{42},\\ M_C &= B_c - 1 - \sum_{e\in\mathcal{E}_c\setminus\{1\}}\rho^e - 2\rho^{1063},\end{aligned}$$

$$M_{F,\mathrm{native}} = 4 + \mu_c(\rho^{4393}+\rho^{5495}) + \sum_{j=0}^{9}(\rho^{1013+j}+\rho^{1065+j}) + \rho^{2195}+\rho^{3296}+Z_c.$$

Use $\rho = 2^{121}$, $B_c = \rho^{8801}$ and the four ports specified earlier. These are bounded finite sums of fixed integers. The source proof checks carry bounds, clause decoding, alignment, mask populations, and the controller for the actual packed rank; they are not new claims established merely by rewriting the recipe here.

References

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[2] J. P. Jones, *Universal Diophantine equation*, Journal of Symbolic Logic **47** (1982), no. 3, 549–571. https://doi.org/10.2307/2273588. The repository's Jones interfaces and explicit finalizer are preserved with their own source notices.

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[4] Z.-W. Sun, *Ten unknowns for Hilbert's tenth problem over the integers*, preprint, arXiv:2609.01594v2, 2 September 2026. https://arxiv.org/html/2609.01594v2. The ten-integer theorem and its number-theoretic finalizer are attributed to Sun; they are not new results of this article.